

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

AU/ME-4002-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Fluid Mechanics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) All questions carry equal marks.
 iii) Assume data suitably.

1. a) Define with mathematical expression: 7
 - i) Uniform and Non-uniform flow
 - ii) Steady and unsteady flow
 - iii) One, two and three dimensional flow
- b) State Newton's law of viscosity. Explain the importance of viscosity in fluid motion. 7

2. a) Explain the following terms: 7
 - i) Path line
 - ii) Streak line
 - iii) Stream line
 - iv) Equipotential line
- b) Derive Euler's equation for flow along stream line and deduce the Bernoulli's Equation for the same. 7

AU/ME-4002-CBGS

PTO

[2]

3. a) The inlet and throat diameters of horizontal venturimeter are 30cm and 10cm respectively. The liquid flowing through the meter is water. The pressure intensity at inlet is 13.734 kN/m² while vacuum pressure head at the throat is 37cm of mercury. Find the rate of flow. Assume that 4% of the differential heads is lost between the inlet and outlet. Find also value of Cd for venturimeter. 7
- b) Derive an expression for continuity equation. 7
4. a) Discharge Q of a centrifugal pump can be assumed to be dependent on density of liquid ρ , viscosity of liquid μ , pressure, impeller diameter D and speed N in R.P.M. Using Buckingham π -Theorem, show that: 7
$$Q = ND^3 \phi \left[\frac{gH}{N^2 D^2}, \frac{V}{ND^2} \right]$$
- b) What is drag and lift? Explain different types of drag on immersed body. 7
5. a) Find the head lost due to friction in a pipe of diameter of 300mm and length 50m, through which water is flowing at a velocity of 3m/s using Darcy equation and Chezy's formula for which L = 60. Given kinematic viscosity 0.01 stores. 7
- b) Derive an expression of velocity and shear stress distribution for laminar flow between fixed parallel plates. 7
6. a) Define Reynold's Number? What is its significance? List the characteristics of Laminar flow. 7
- b) Derive an expression for Total Pressure Force and depth of centre of pressure for a vertical surface submerged in water. 7

AU/ME-4002-CBGS

Contd...

[3]

7. a) Explain the different types of similitude. 7
- b) Define the following terms and write relevant equations for the same: 7
 - i) Stagnation temperature
 - ii) Stagnation pressure

8. Write notes on: 14
 - i) Momentum Thickness
 - ii) Sub sonic and Super sonic flow
 - iii) Buoyancy
 - iv) Surface Tension
